Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.2 – Current & Voltage

Unit 2: Electricity & Energy

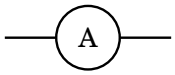
The question: *you can draw the circuit now — but how do you know how much electricity is flowing, and how hard the cell is pushing it?* Two meters, two different jobs. Mix them up and you measure the wrong thing.

Learning intentions

All students will state what **current** and **voltage** each measure, and connect an ammeter and a voltmeter correctly.
Most students will explain *why* each meter goes where it goes.
Some students will calculate **charge** from current and time.

Part 1 — Current

Current is how much charge flows past a point *every second*. More current, more charge moving — think of it as how much "electricity traffic" is going round the loop. Current is measured in **amps (A)**, with an **ammeter**.

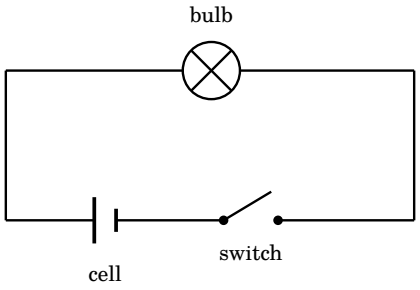


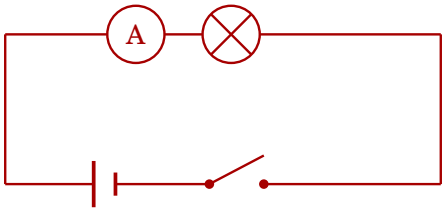
Ammeter

measures current — goes IN the loop, in series

An ammeter measures the current *through* a component, so **all** of that current has to pass through the meter too. That only works if the ammeter is wired into the loop itself, like another component.

1) **Draw** an ammeter into the circuit below, so it reads the current through the bulb.

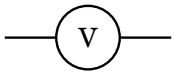




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Part 2 — Voltage

Voltage (or **potential difference**) is how hard the cell pushes each bit of charge round the loop — the energy given to the charge at the cell, and taken back off it at the bulb. Voltage is measured in **volts (V)**, with a **voltmeter**.

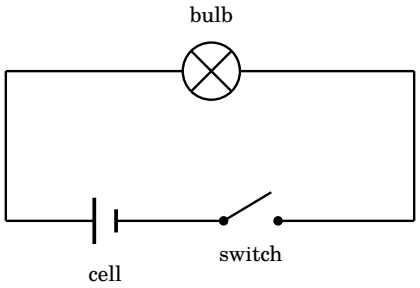


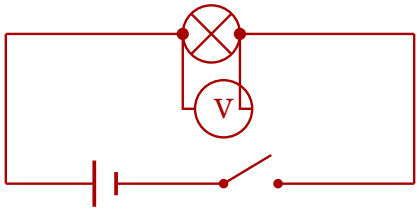
Voltmeter
measures voltage —
goes ACROSS a
component, in parallel

A voltmeter compares two points — before and after a component — so it connects to *both* ends of that component, alongside the main loop, not inside it.

Word bank: ammeter → series → in the loop | voltmeter → parallel → across a component

2) **Draw** a voltmeter across the bulb in the same circuit, so it reads the voltage across it.





Build it

Build the circuit with a real ammeter in series and a real voltmeter in parallel across the bulb. Close the switch and record both readings.

Current: _____ A **Voltage:** _____ V

Check: both meters should settle on a steady, non-zero reading the moment the switch closes. A reading of zero on the ammeter with the bulb lit means the ammeter is not actually in the loop — trace its wires.

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Most students — why each meter goes where it goes

3) **Explain** why an ammeter *has* to be in series, and a voltmeter *has* to be in parallel. Use the word **all** in your ammeter answer and the word **two** in your voltmeter answer.

_____

_____

_____

Model answer: *Ammeter: it measures the current through a component, so **all** of that current must pass through the meter — which only happens if it is wired into the loop itself. Voltmeter: it compares the energy at **two** points, so it needs a connection to both of them, alongside the component rather than inside the loop.*

Some students — charge

Current is charge flowing per second, so: **charge** = **current** × **time**, or $Q = I \times t$. Charge is measured in **coulombs (C)**.

4) **Calculate** the charge that flows through a bulb in 20 seconds, if the current through it is 0.3 A.

_____

Model answer: $Q = I \times t = 0.3 \times 20 = 6\,C$.

Part 3 — On your own

Here is a circuit with **two** bulbs, one after another in a single loop (the one you drew for homework last lesson).

5) **Draw** the circuit diagram, then add an ammeter and a voltmeter so that together they measure the current through, and the voltage across, only the **first** bulb.

Two bulbs in series; an ammeter in the loop immediately before or after the first bulb; a voltmeter connected across the first bulb only (not the second).

HOMEWORK

Find a battery at home (a AA, a remote-control battery, anything with numbers printed on it). Write down the voltage printed on it, then explain in one sentence what that number tells you, now that you know what voltage means.